Data Science

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Course Outline:

- Data Collection
- Data Cleaning & Preparation
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Code

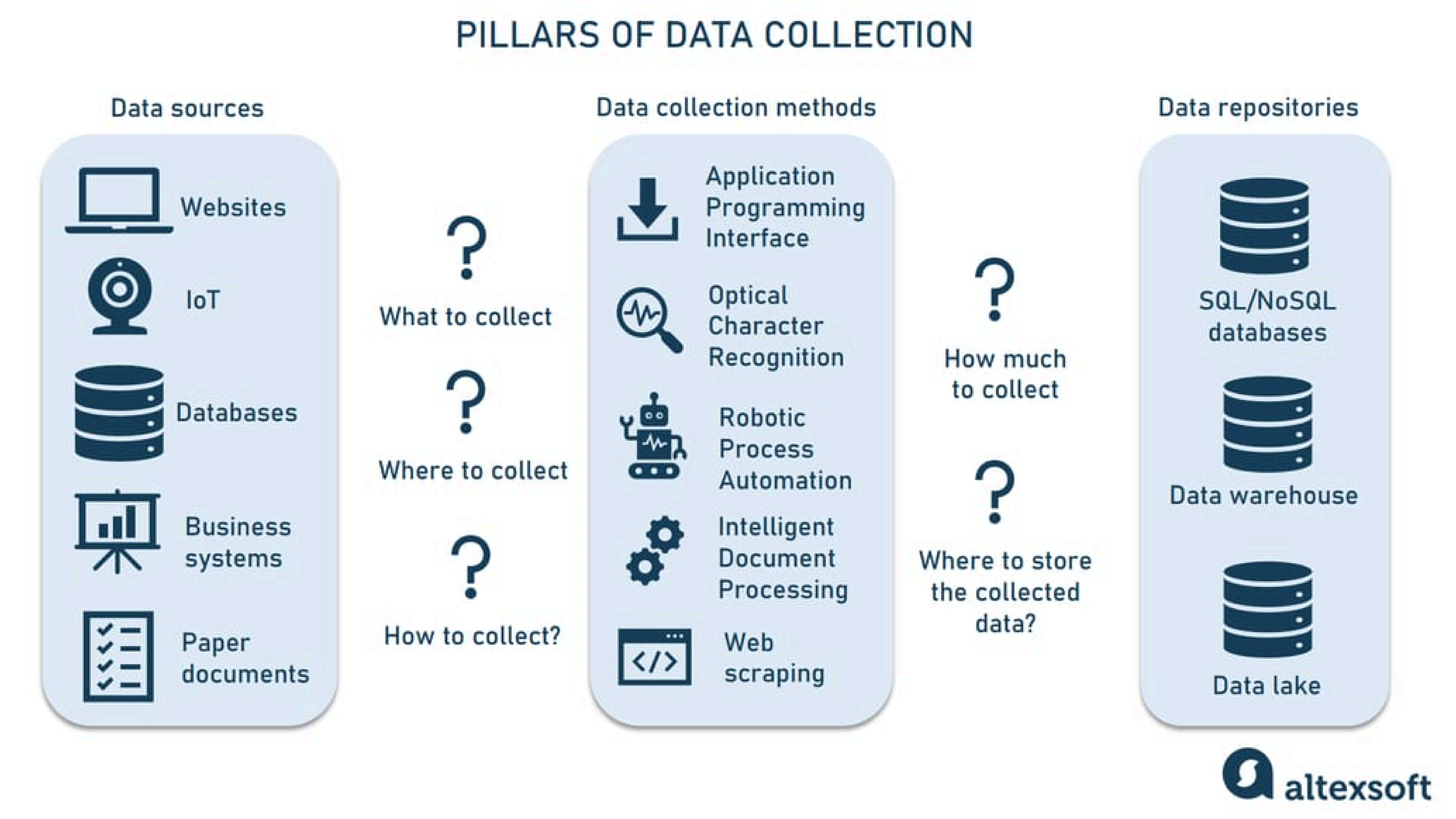
Problem & Data

- What is the business goal or research question?
- What are the variables in the data and what do they represent?
- What types of data (numerical, categorical, text, etc.) do you have?
- Are there any known data quality issues or limitations?
- Are there any domain-specific concerns or restrictions?

What is Data Collection?
When is Data Collected?

Types of Data Sources

1. Structured Data
2. Unstructured Data
3. Static vs Streaming Data
4. Primary vs Secondary Data


PILLARS OF DATA COLLECTION
Data sources
Websites
IoT
Databases
Business systems
Paper documents
What to collect
Where to collect
How to collect?
Data collection methods
Application Programming Interface
Optical Character Recognition
Robotic Process Automation
Intelligent Document Processing
Web scraping
How much to collect
Where to store the collected data?
Data repositories
SQL/NoSQL databases
Data warehouse
Data lake
altexsoft

Key Challenges:

Data Provenance
Data Labelling
Missing/Noisy Data
Ethical Concerns
Sampling Bias

Best Practices:

Define Clear Objectives
Document Metadata
Validate Early

Mistakes

Ignoring Licensing
Silent Failures
No Versioning

Best Practices:

Automate Collection
Store Raw Data Separately
Log Errors
Respect Ethics

Most Important Libraries

```
[ ]: import numpy as np                    # For numerical operations
     import pandas as pd                   # For data manipulation
     import seaborn as sns                 # For loading sample datasets and plotting
     import matplotlib.pyplot as plt       # For visualization
     from sklearn.model_selection
     from sklearn.preprocessing
```

Data Cleaning & Preparation

Key Goals

Remove Noise
Handle Missing Data
Standardize Formats
Structuring

When is it Used?

Immediately after data collection.
Before exploratory analysis or modeling.
When integrating data from multiple sources

Why is it Critical?

Model Performance
Efficiency
Reproducibility

Key Steps in Data Cleaning & Preparation

1- Handle Missing Data:

Identify: Use summary statistics

Strategies:

- **Delete: Drop rows if they're few, columns if it has a lot**
- **Impute: Replace with mean/median (numeric) or mode (categorical).**
- **Predict: Use machine learning (like: KNN imputation).**

Address	City	Subject	Marks	Rank	Grade
123 Main St	Los Angeles	Math	85.0	2	B
456 Oak Ave	New York	English	92.0	1	A
789 Pine Ln	Houston	Science	78.0	4	C
101 Elm St	Los Angeles	Math	89.0	3	B
NaN	Miami	History	NaN	8	D
22 Maple Rd	NaN	Math	95.0	1	A
4 Cedar Blvd	Houston	Science	80.0	5	C
555 Birch Dr	New York	English	88.0	3	B

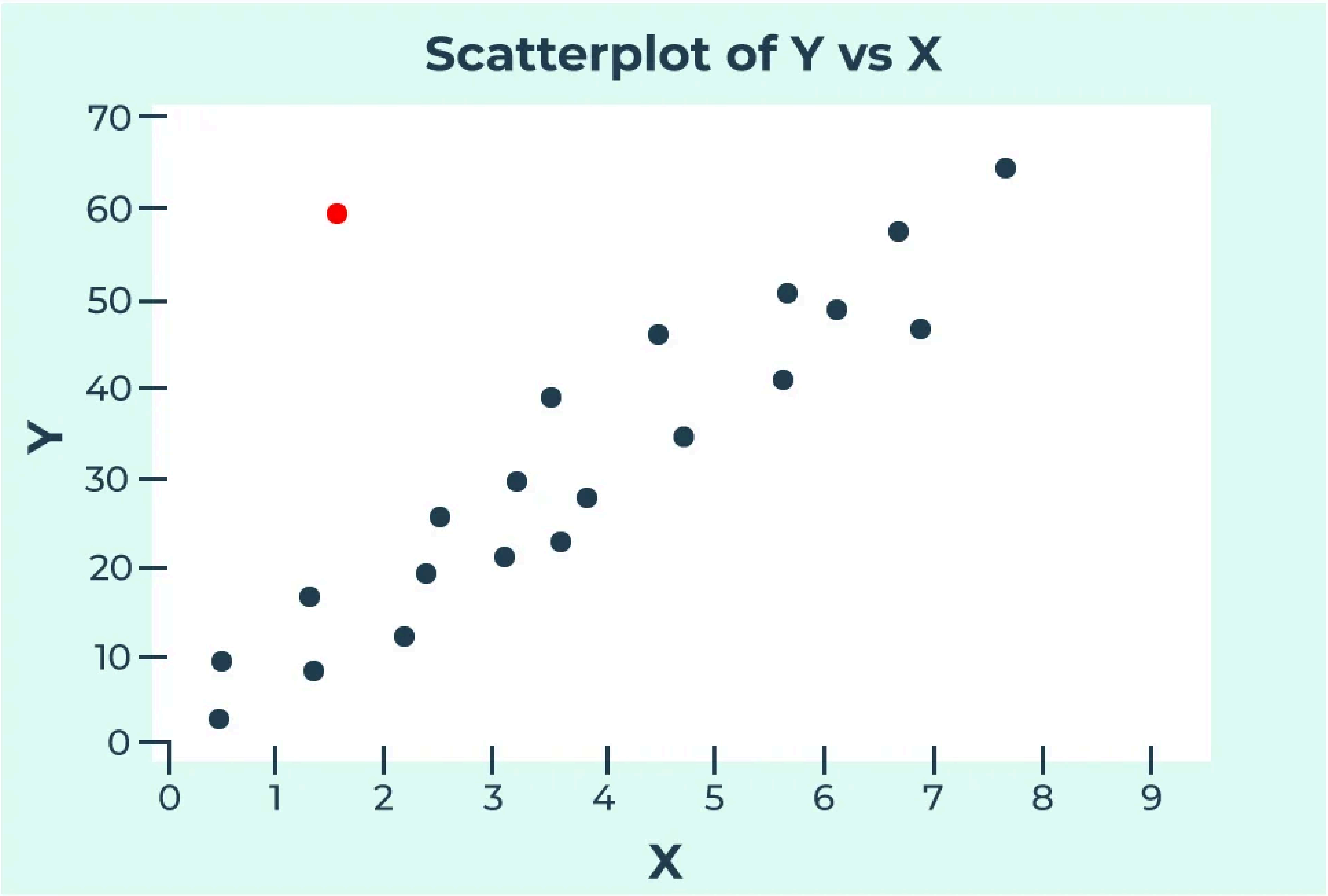
(MCAR) | (MAR) | (MNAR)

Key Steps in Data Cleaning & Preparation

2. Remove Duplicates:
    - Use df.duplicated() to find exact or near-duplicates.

3. Fix Structural Errors:
    - Typos (like: "Fmale" → "Female").
    - Inconsistent formats (e.g., "2023-01-01" vs. "01/01/2023").

Handling Outliers



Handling Outliers

but how to detect them? 3 ways

(a) Using Z-Scores (Standardization)

$$Z = \frac{(X - \mu)}{\sigma}$$

- X = data point
- μ = mean
- σ = standard deviation

If Z=3, the data point is 3 stadard deviations away so anything more than that its considered an outlier

Handling Outliers

<u>**but how to detect them? 3 ways**</u>

(b) Using Interquartile Range (IQR)

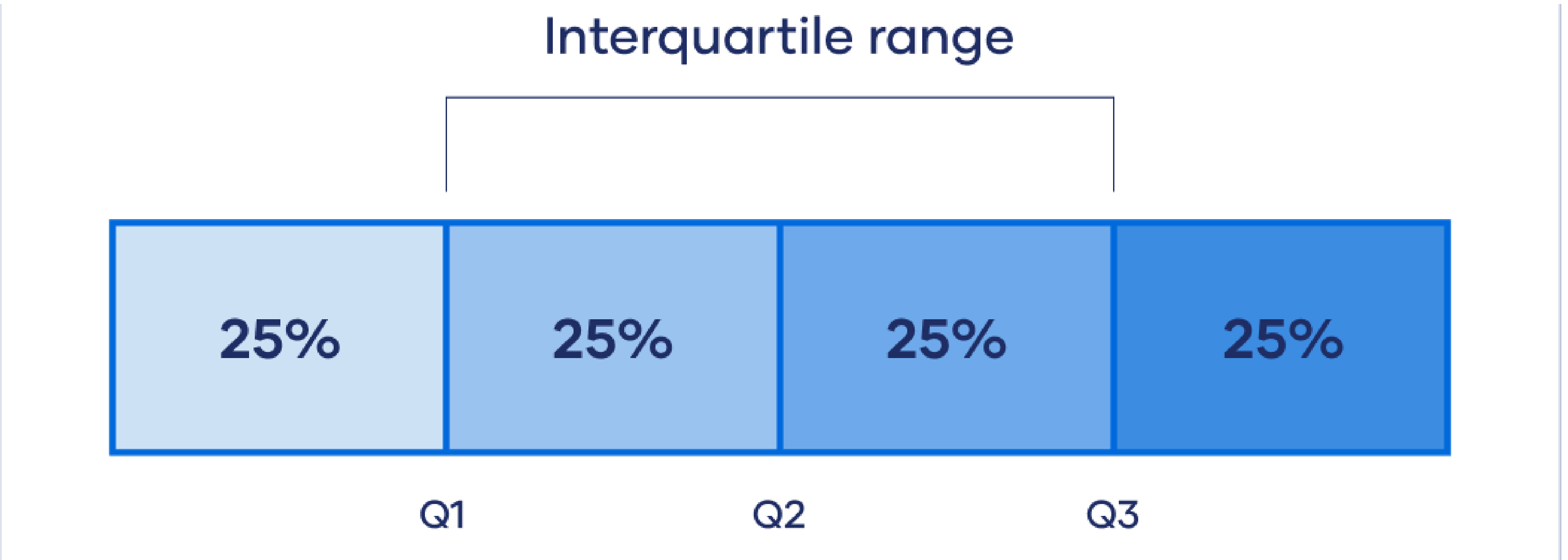
$$IQR = Q3 - Q1$$

Outliers are defined as:

$$\text{Lower bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper bound} = Q3 + 1.5 \times IQR$$

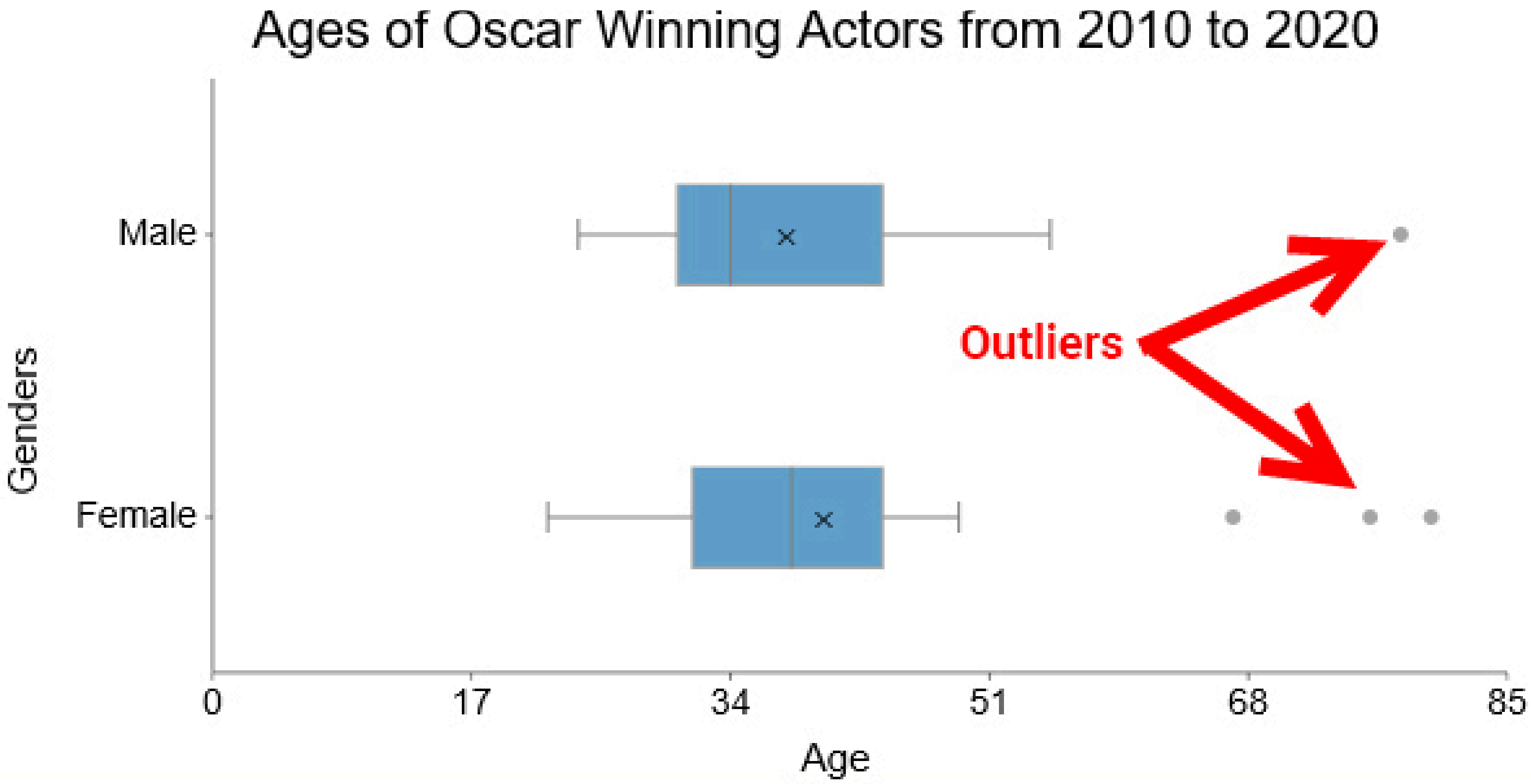
If a value falls outside this range, it is considered an outlier



Handling Outliers

<u>but how to detect them? 3 ways</u>

(c) Using Visualization (Boxplots & Scatterplots)



https://chartexpo.com/blog/box-plot-outliers#

Handling Outliers

but how to resolve them?

(a) Remove Outliers

```
filtered_data = data[(data >= lower_bound) & (data <= upper_bound)]
```

(b) Cap/Floor

```
data = np.clip(data, lower_bound, upper_bound)
```

(c) Transform Data (Log, Square Root, Box-Cox)

```
log_data = np.log(data + 1)
```

(d) Investigate Contextually

5. Encode Categorical Variables:
    - Ordinal: Label encoding ("Low", "Medium", "High" → 0, 1, 2).
    - Nominal: One-hot encoding (Red", "Blue" → separate columns).
6. Validate Data Types:
    - Ensure numeric columns are float/int, dates are datetime, etc.

7. **Splitting dataset into training, validation and testing set**

Mistakes

Deleting Data Blindly
Ignoring Data Distribution
Over-Engineering

Best Practices

Document Changes
Version Control
Automate Pipelines
Validate Post-Cleaning

EDA

Exploratory Data Analysis

Why Perform EDA?

Identify data distributions, outliers, and missing values (even after cleaning).

Discover relationships between variables (like: correlation, causation).

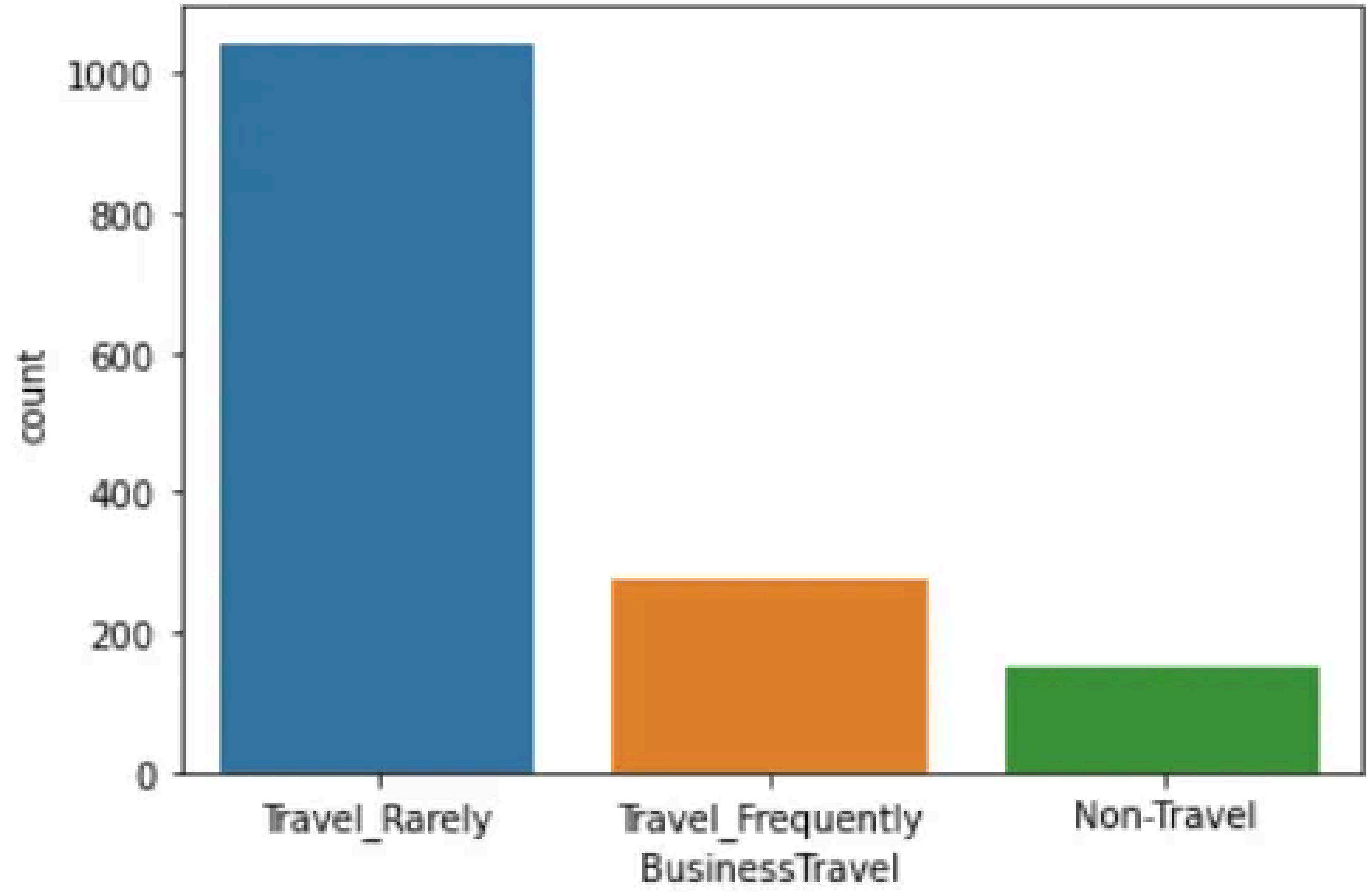
Inform feature engineering and model selection (skewed data → log transform).

Key Techniques in EDA

Univariate Analysis

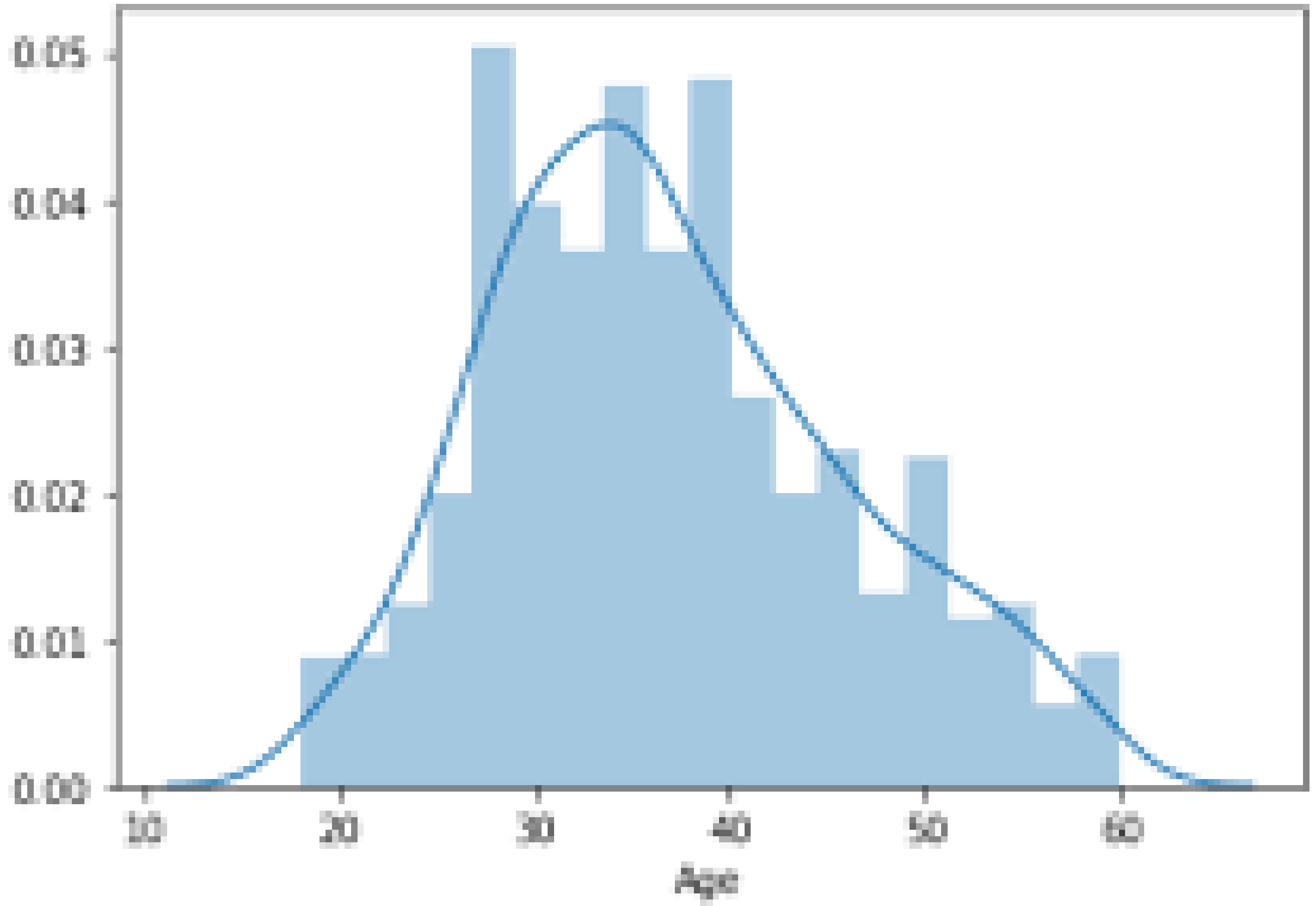
```python
sns.countplot(data['BusinessTravel'])
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x1f68c55c748>
```




```python
#Distributiong of Age variable
sns.distplot(data['Age'])
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x1f68a445a20>
```

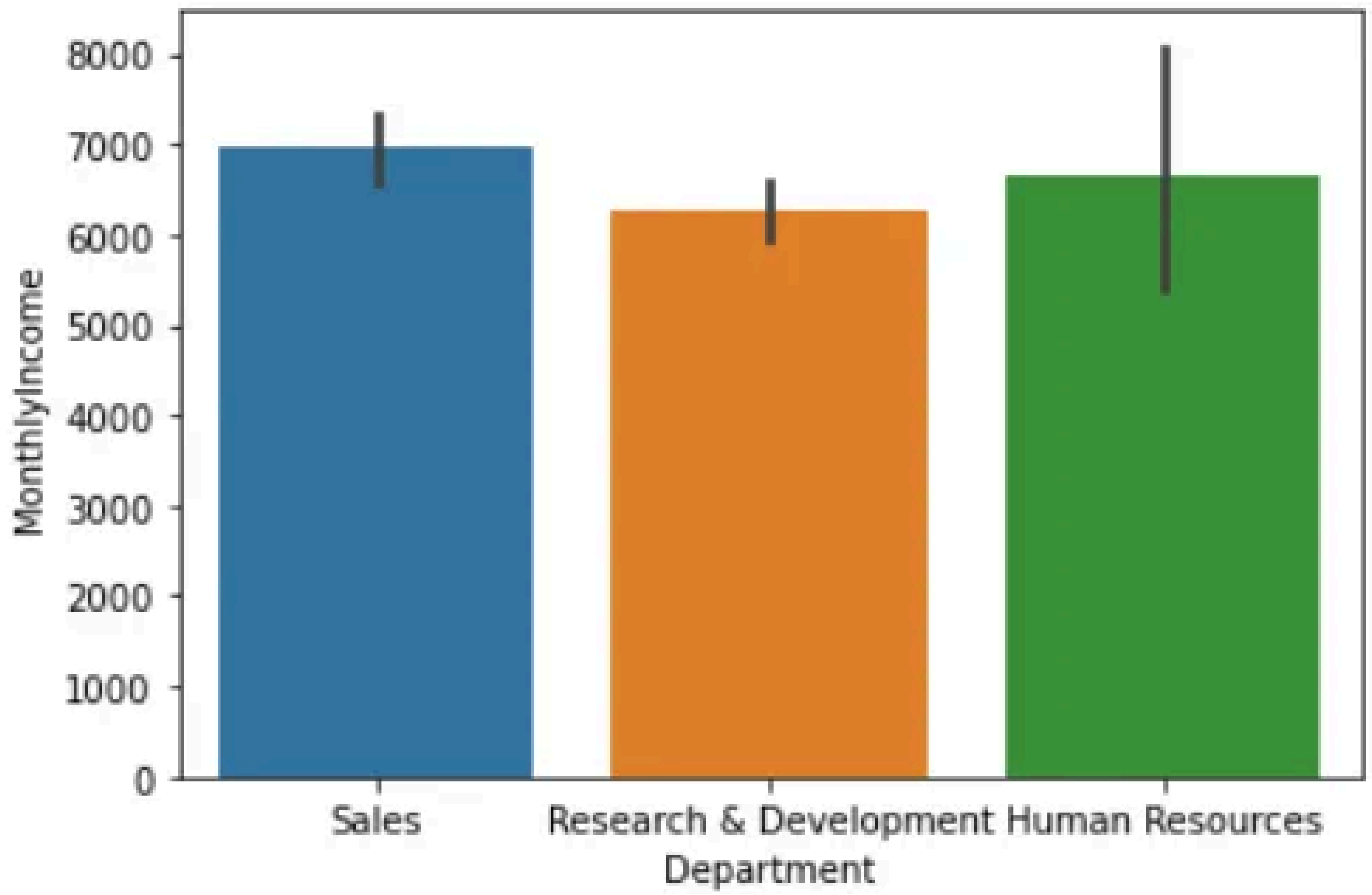


Key Techniques in EDA

Bivariate/Multivariate Analysis

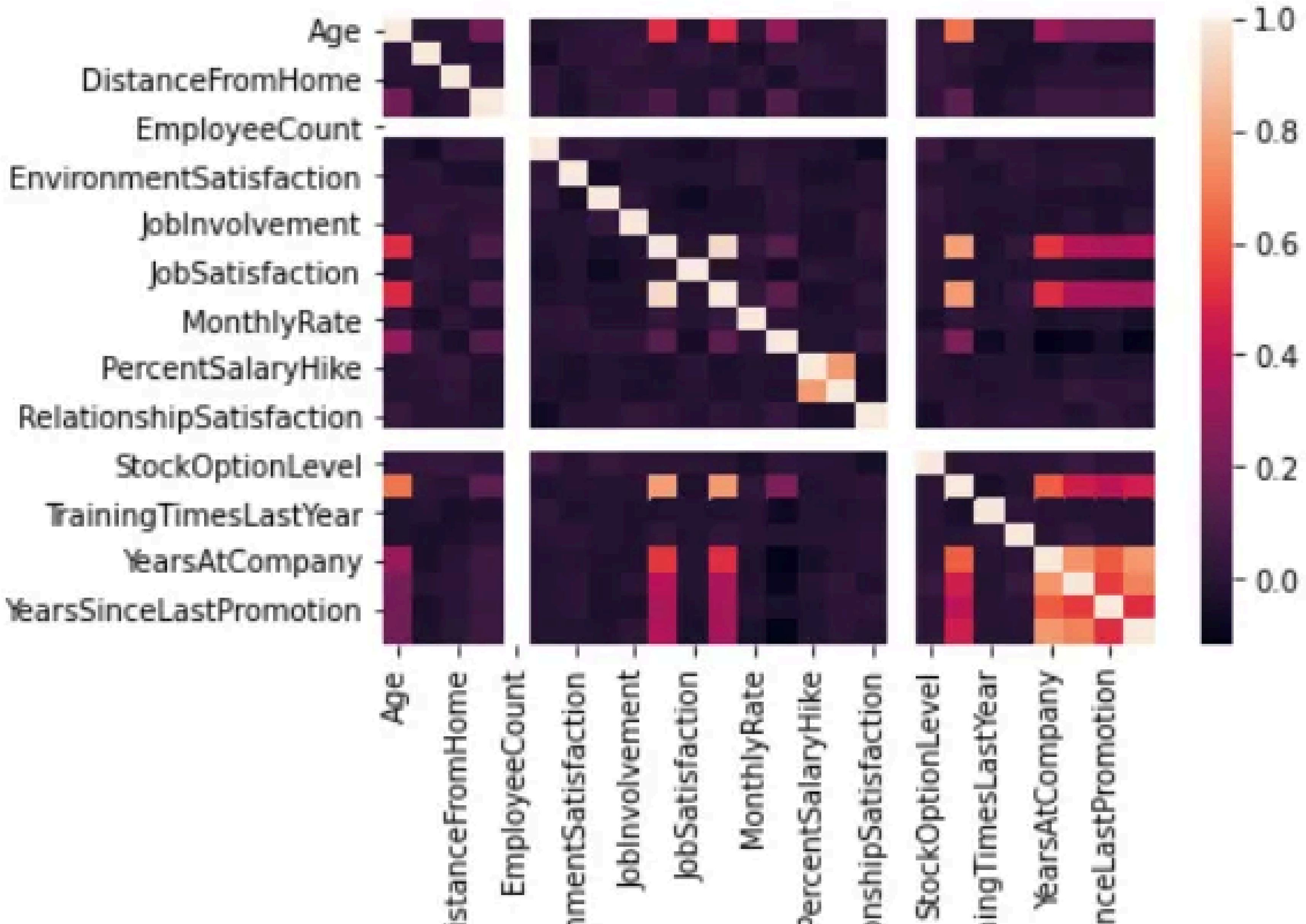
```
sns.barplot(data['Department'], data['MonthlyIncome'])
```

<matplotlib.axes._subplots.AxesSubplot at 0x1dd99744668>



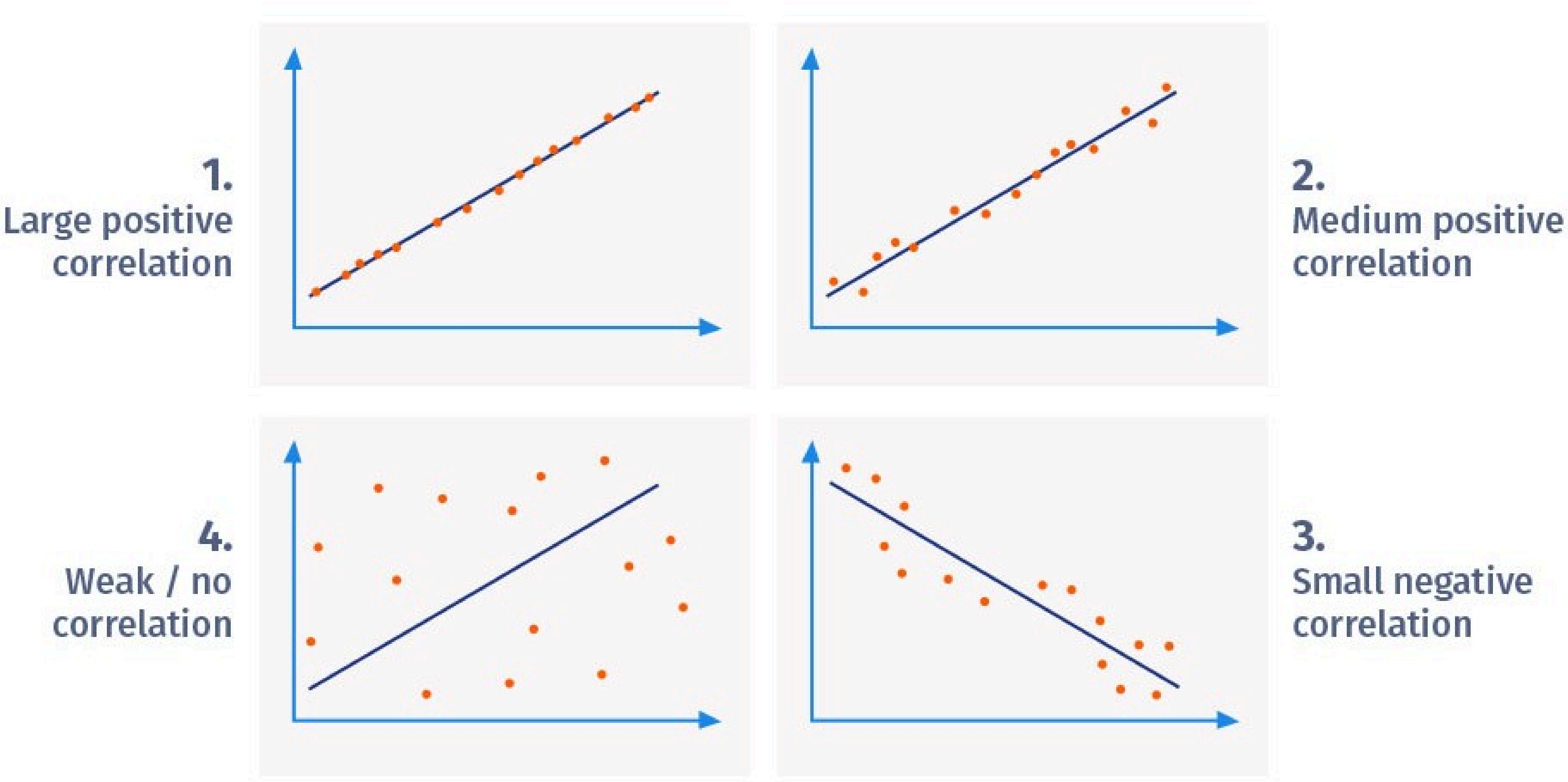

```
sns.heatmap(data.corr())
```

<matplotlib.axes._subplots.AxesSubplot at 0x18b97ed6b00>



Key Techniques in EDA

Correlation Analysis



Mistakes

Ignoring Skewed Distributions
Overplotting
Misinterpreting Correlation

Best Practices

Start Simple
Layer Information
Document Observationsg

Feature Engineering

raw data ⟶ features ⟶ modeling

↑

feature engineering

Feature Engineering

<u>Feature..</u>

Transformation **Creation** **Extraction**

Selection **Scaling**

Feature Engineering

Feature Transformation

Normalization(Min-Max Scaling)

Standardization

Encoding Categorical Features

Mathmatical

Feature Engineering

<u>Feature Selection</u>

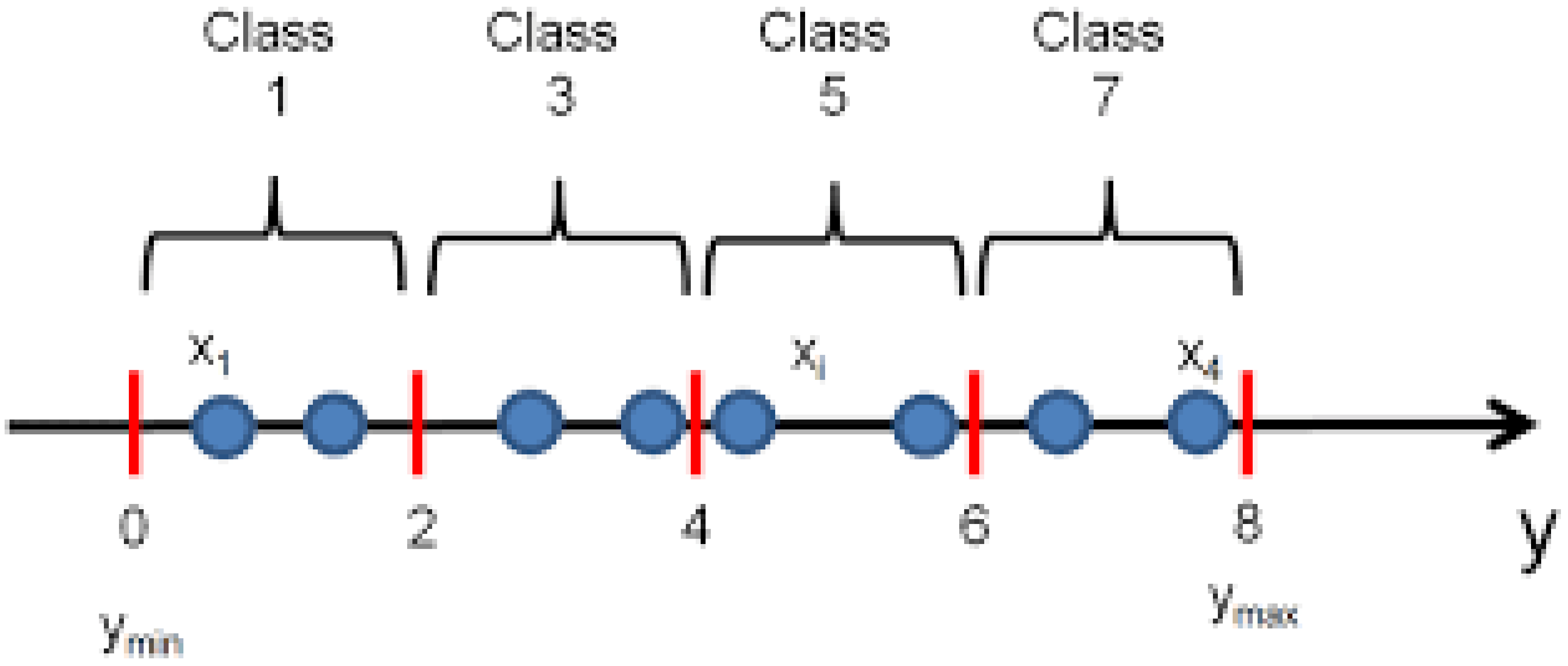
Relation with Target vairable

Using Algorthims

Testing

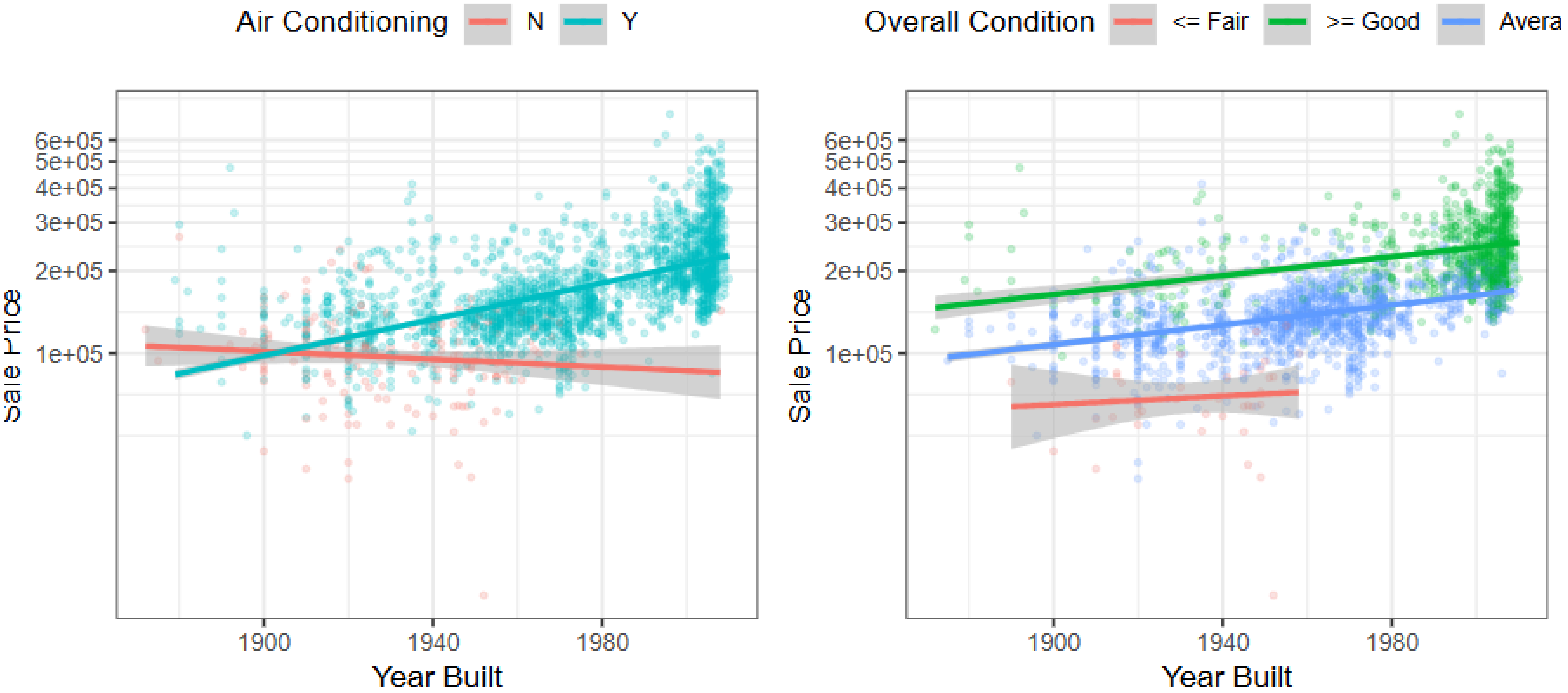
Key Techniques in Feature Engineering

Binning/Discretization



Key Techniques in Feature Engineering

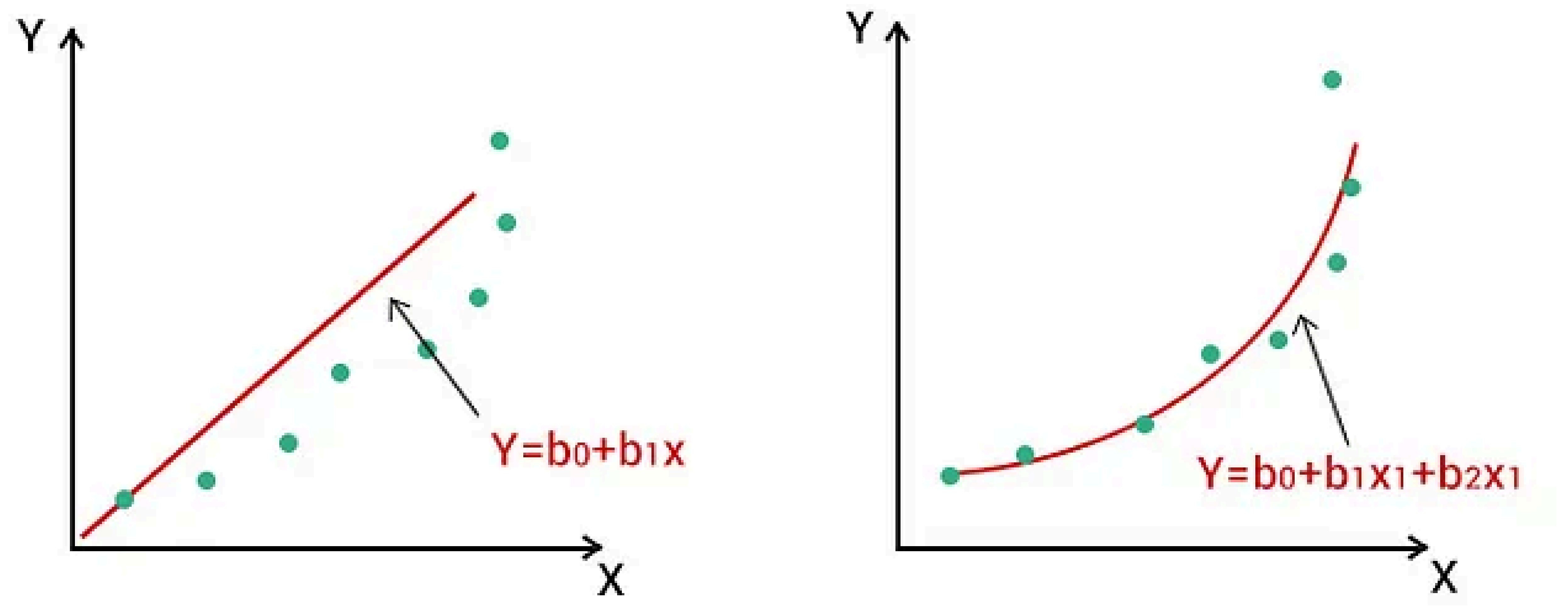
Interaction Terms/



Key Techniques in Feature Engineering

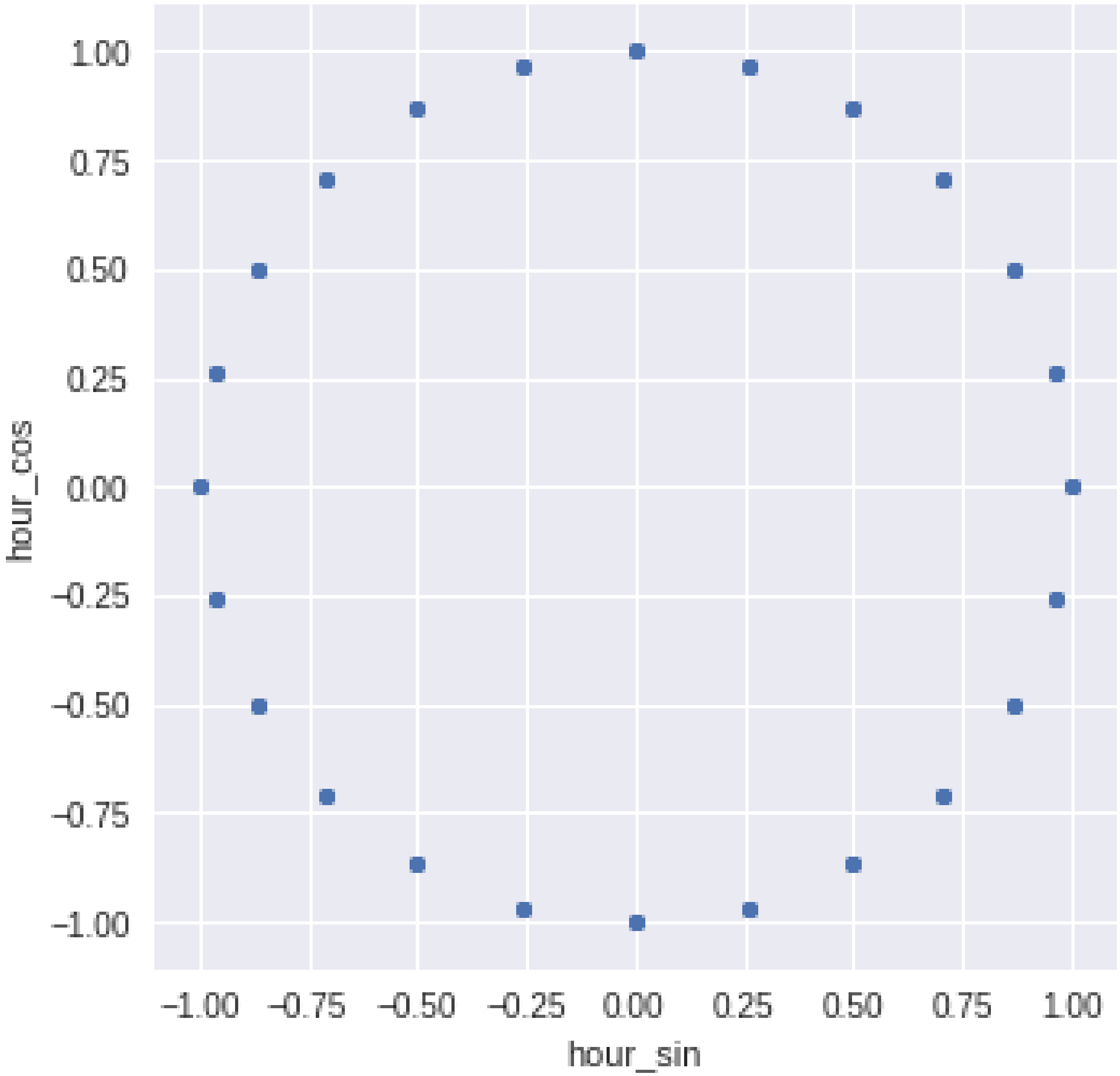
Polynomial Features

Polynomial Feature Transform in Machine Learning



Key Techniques in Feature Engineering

Encoding Cyclical Features



Mistakes

Over-Engineering
Ignoring Domain Context
Data Leakage

Best Practices

Start Simple
Validate with EDA
Use Domain Knowledge